

A Multimodal LSTM-CNN Based Approach to Audiovisual Emotion Recognition

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Abstract—Automatically recognizing human emotions from spontaneous and non-prototypical real-life data remains a significant challenge in affective computing. This dissertation investigates the application of Long Short-Term Memory (LSTM) networks for speech emotion recognition and Convolutional Neural Networks (CNN) for facial emotion recognition within audiovisual human-computer interaction. We utilized the RAVDESS datasets for audio and the FER-2013 dataset for facial image recognition. Our goal was to integrate these models into a comprehensive multimodal system that could accurately recognize emotions from both audio and visual inputs. While the LSTM model effectively captured the temporal dynamics of emotions in speech, leveraging word-level audio features to achieve superior recognition performance, the CNN model processed facial movements to recognize emotions from images. However, we encountered significant challenges in integrating these models due to differences in input sizes and structures across the datasets. These challenges prevented the full realization of a combined audiovisual emotion recognition system. Nonetheless, our work demonstrates substantial progress in speech emotion recognition, and we hope that future research will overcome these integration challenges to develop a robust multimodal emotion recognition system.

Keywords— Emotion Recognition, Long Short-Term Memory, Affective Computing, Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN), Multimodal Systems, Speech Emotion Recognition, Human-Computer Interaction (HCI), Audiovisual Integration.

I. INTRODUCTION

In recent years, the intersection of artificial intelligence and human emotion has emerged as a captivating frontier, with profound implications for both technology and society. Automatic Speech Recognition (ASR) and Speech Emotion Recognition (SER) are pivotal areas where advancements have not only enabled machines to transcribe and understand spoken language but also to discern the subtle emotional nuances embedded within human speech. However, as the field matures, it becomes increasingly evident that speech alone does not fully encapsulate the richness of emotional communication. Human emotions are complex, multifaceted phenomena that are also expressed through visual cues such as facial expressions, gestures, and body language.

This dissertation delves into the realm of multimodal emotion recognition, specifically focusing on the integration of audio

and visual information. The advent of deep learning techniques, particularly Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNN), has opened new avenues for processing sequential and visual data. LSTM networks are particularly suitable for capturing temporal dependencies in audio streams, while CNNs excel in recognizing patterns in visual data, such as facial expressions. By harnessing the complementary strengths of these modalities, this research aims to enhance the accuracy and robustness of emotion recognition systems.

Traditional approaches in SER have predominantly relied on unimodal data, utilizing features extracted solely from audio signals such as Mel-frequency cepstral coefficients (MFCC) or spectrograms. While these methods have achieved notable success, they inherently overlook crucial contextual cues provided by visual information. Conversely, multimodal frameworks integrate audiovisual inputs, leveraging the synergies between different modalities to achieve more comprehensive and nuanced emotional understanding. Such approaches have shown promising results, outperforming their unimodal counterparts by a significant margin.

In this research, we employ LSTM networks for processing MFCC features and mel-spectrograms extracted from audio signals, utilizing a range of data augmentation techniques, including noise injection, stretching, and pitch modulation, to improve model resilience and generalization. For the visual modality, we use CNNs to analyze facial expressions from the FER-2013 dataset, incorporating callback functions before training the model to optimize learning processes. Despite the challenges encountered in integrating these models due to differences in input sizes and structures, this work makes significant strides in multimodal emotion recognition.

By advancing our understanding of how emotions manifest across different modalities, this dissertation seeks to contribute to the broader discourse on human-computer interaction. The ability to accurately recognize and respond to emotional cues not only enhances the functionality of AI systems but also holds transformative potential in fields ranging from healthcare and education to virtual assistants and entertainment. Ultimately, by bridging the gap between audio and visual emotion recognition, this research aims to pave the way for more empathetic and intuitive artificial intelligence systems, capable of engaging with users in a manner that mirrors human understanding.

II. LITERATURE REVIEW

The rise of affective computing has spurred significant research into multimodal emotion recognition, which combines audio and visual cues to better understand human emotions. This review highlights key studies in the field, focusing on the use of Long Short-Term Memory (LSTM) networks for processing audio and Convolutional Neural Networks (CNN) for visual data, and addresses the challenges in integrating these two modalities.

A. Unimodal Emotion Recognition Approaches

Early research in Speech Emotion Recognition (SER) primarily relied on unimodal approaches, focusing on extracting features from audio signals to predict emotional states. Features such as Mel-frequency cepstral coefficients (MFCC), pitch, energy, and formants were extensively used in conjunction with machine learning classifiers like Support Vector Machines (SVMs) and Hidden Markov Models (HMMs) (Cowie et al., 2001; Schuller et al., 2004). Although these methods achieved reasonable success, they were limited by their reliance on handcrafted features and the inability to capture temporal dependencies effectively.

The introduction of deep learning techniques, particularly LSTM networks, revolutionized SER by enabling the modeling of long-range temporal dependencies in speech data. LSTMs, a type of Recurrent Neural Network (RNN), were designed to overcome the vanishing gradient problem inherent in traditional RNNs, making them well-suited for tasks involving sequential data (Hochreiter & Schmidhuber, 1997). Research by Trigeorgis et al. (2016) demonstrated that LSTM networks could outperform traditional approaches by automatically learning relevant features from raw audio signals and capturing the temporal evolution of emotions.

B. Visual Emotion Recognition

Parallel to advancements in audio-based emotion recognition, significant progress has been made in visual emotion recognition. The use of CNNs, which are adept at recognizing patterns in image data, has become the dominant approach for analyzing facial expressions—a primary visual indicator of emotions. Early works by LeCun et al. (1998) laid the foundation for CNNs, which have since been refined and applied to various image recognition tasks, including facial emotion recognition (Krizhevsky et al., 2012). Datasets such as FER-2013 (Goodfellow et al., 2013) have been instrumental in advancing the field of visual emotion recognition. CNN-based models have been shown to achieve high accuracy in identifying basic emotions such as happiness, sadness, and anger by analyzing facial expressions. However, challenges remain, particularly in recognizing subtle or complex emotions and dealing with variations in lighting, occlusions, and head poses (Mollahosseini et al., 2016). Recent studies have focused on improving the robustness of CNN models through techniques such as data augmentation, the use of attention mechanisms, and the incorporation of temporal information (Zhang et al., 2017).

C. Multimodal Emotion Recognition

Given the limitations of unimodal approaches, there has been a growing interest in developing multimodal emotion

recognition systems that integrate both audio and visual data. The rationale behind multimodal approaches is that human emotions are inherently multimodal, with different cues providing complementary information. Early attempts at multimodal emotion recognition often combined handcrafted features from both modalities using traditional machine learning classifiers (Busso et al., 2004; Zeng et al., 2009).

With the advent of deep learning, more sophisticated architectures have been proposed. Poria et al. (2017) introduced a multimodal system that utilized LSTMs for both audio and textual data, combined with CNNs for visual data. Their work demonstrated that multimodal systems could significantly outperform unimodal systems, particularly in scenarios where one modality was noisy or unreliable. However, they also highlighted the challenges of integrating different modalities, such as aligning audio and visual data streams and managing the varying input sizes and structures across datasets.

D. Fusion Techniques in Multimodal Emotion Recognition

Integrating audio and visual modalities requires effective fusion techniques to combine the information from both channels in a manner that enhances the overall emotion recognition accuracy. The two primary approaches to fusion are feature-level fusion and decision-level fusion. Feature-level fusion involves combining features extracted from each modality before feeding them into a classifier, allowing for the modeling of interactions between the features of different modalities. However, this approach can lead to high-dimensional feature spaces, making the model prone to overfitting and increasing computational complexity. In contrast, decision-level fusion combines the outputs (predictions) of separate classifiers for each modality. This approach is often more robust to modality-specific noise and can be more easily implemented when the modalities are asynchronous or have different sampling rates. Recent advancements in deep learning have led to the development of hybrid fusion methods, where intermediate layers of the neural networks are combined. This can include methods like attention mechanisms, which dynamically weigh the contributions of each modality.

E. Role of Attention Mechanisms

Attention mechanisms have become increasingly important in the field of multimodal emotion recognition. These mechanisms allow models to focus on the most relevant parts of the input data, enhancing performance by filtering out irrelevant information. Self-attention mechanisms enable models to assign different weights to different parts of a single modality input, such as different frames in a video or different segments of an audio signal. This technique has been particularly successful in dealing with long-range dependencies and enhancing the temporal modeling capabilities of LSTMs. Studies have shown that self-attention can improve the accuracy of emotion recognition by allowing the model to focus on emotionally salient parts of the input data. Cross-modal attention, on the other hand, involves using one modality's features to guide the attention over another modality. For example, visual features might help the model focus on specific segments of the audio that are more likely to contain emotional content. This approach has been shown

to enhance the synergy between modalities, leading to more accurate emotion recognition.

This study builds on these foundational works by exploring the integration of LSTM networks for audio and CNNs for visual emotion recognition. While previous studies have shown the potential of these models individually, integrating them into a cohesive system remains a challenge. One of the primary difficulties lies in handling the discrepancies between audio and visual data, such as differences in data sampling rates and input dimensions. Additionally, the variability in datasets, as seen in the RAVDESS, CREMA-D, and FER-2013 datasets, further complicates the integration process.

III. METHODOLOGY

A. Datasets Description

For the audio component:

RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song): The RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song) dataset, developed by Livingstone and Russo in 2018, is a critical resource in the field of emotion recognition. It provides high-quality recordings of 24 actors (12 male and 12 female) expressing a range of eight emotions: neutral, calm, happy, sad, angry, fearful, disgusted, and surprised, with varying levels of intensity. The dataset includes both speech and song modalities, with a focus on speech in this research. The recordings are in North American English and are provided as 16-bit, 48kHz WAV audio files, ensuring consistency and clarity. RAVDESS is particularly well-suited for speech emotion recognition tasks due to its emotional diversity and the controlled environment in which it was recorded. However, the controlled nature of the dataset and the actors' exaggerated emotional expressions may limit the model's ability to generalize to real-world scenarios, potentially leading to overfitting.

For the video component:

FER-2013 (Facial Expression Recognition 2013): Dataset, introduced during the ICML 2013 conference, is widely used for facial emotion recognition. It contains 35,887 grayscale images, each with a resolution of 48x48 pixels, depicting facial expressions categorized into seven emotions: anger, disgust, fear, happiness, sadness, surprise, and neutrality. The dataset is split into 28,709 training images, 3,589 validation images, and 3,589 test images, making it a robust tool for training and evaluating deep learning models. FER-2013's large size and diverse sources make it highly valuable for training Convolutional Neural Networks (CNNs) in facial emotion recognition. However, the low resolution of the images, variability in quality, and an imbalance in the "disgust" category present challenges that may require data augmentation and careful handling to prevent model bias and ensure effective emotion recognition.

These datasets provide a rich foundation for training and evaluating the proposed multimodal approach to audiovisual emotion recognition using LSTM networks for audio and CNNs for visual data. They enable the system to learn and generalize emotional features across different modalities, contributing to advancements in understanding and interpreting human emotions through machine learning.

B. Proposed Model

The following sections detail the architecture and implementation of both models.

Convolutional Neural Network (CNN) for Visual Emotion Recognition

The CNN model is designed to classify facial expressions into seven emotion categories: anger, disgust, fear, happiness, sadness, surprise, and neutrality. The model architecture comprises multiple layers aimed at progressively extracting and refining features from input images.

- **Input Layer:** The model accepts grayscale images with a shape of '(row, col, 1)'. The first convolutional layer applies 32 filters of size '(3x3)', followed by another convolutional layer with 64 filters of the same size. Both layers use the ReLU activation function and apply zero-padding to preserve the spatial dimensions.
- **Batch Normalization and Pooling:** After the initial convolutional layers, a Batch Normalization layer is added to stabilize the learning process. Max Pooling with a '(2x2)' filter is then applied to reduce the spatial dimensions by half, followed by a Dropout layer with a 25% dropout rate to prevent overfitting.

Max pooling reduces the dimensionality of the input by taking the maximum value in a window. For a 2x2 pooling operation:

$$Z_{i,j,k} = \max(\{X_{2i,2j,k}, X_{2i+1,2j,k}, X_{2i,2j+1,k}, X_{2i+1,2j+1,k}\})$$

Where:

- $Z_{i,j,k}$ is the output after max pooling.
- X is the input feature map.
- **Deeper Convolutional Layers:** The model continues with two more convolutional layers, with 128 and 256 filters respectively, both using ReLU activation. These layers also include regularization via an L2 penalty '(0.01)' to further mitigate overfitting. Another Batch Normalization and MaxPooling layer is added, followed by a 25% Dropout layer.
- **Fully Connected Layers:** The output from the convolutional layers is flattened and passed through a fully connected layer with 1024 neurons, using ReLU activation. A Dropout rate of 50% is applied to reduce overfitting. Finally, the output layer uses a SoftMax activation function to classify the input into one of the seven emotion categories.

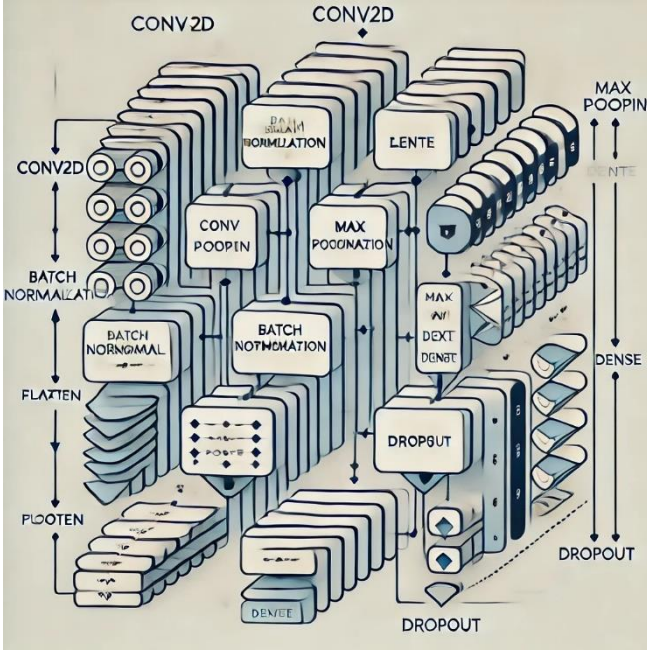


Fig1: A fully connected CNN model

- **Model Compilation:** The model is compiled using the Adam optimizer with a learning rate of 0.0001 and a decay of '1e-6'. The loss function used is categorical cross-entropy, appropriate for multi-class classification, and accuracy is the primary metric for evaluation.

Long Short-Term Memory (LSTM) Network for Speech Emotion Recognition

For the audio modality, we employed an LSTM network to capture the temporal dependencies inherent in speech signals, allowing the model to recognize emotions based on the sequential nature of audio data.

- **Input Layer:** The input to the LSTM model is reshaped to ensure compatibility with Conv1D operations, adapting the audio data into a format suitable for time-distributed processing.
- **Convolutional Layers:** The model starts with a Time_Distributed Conv1D layer that applies 16 filters of size '(3)', with ReLU activation and 'same' padding, allowing the temporal features to be extracted from the sequential data. This is followed by Batch Normalization to stabilize learning across time steps.

The operation of a convolutional layer can be represented as:

$$Z_{i,j,k} = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} X_{i+m,j+n,c} \cdot W_{m,n,c,k} + b_k$$

Where:

- $Z_{i,j,k}$ is the output of the convolution at position (i, j) for filter k.
- X is the input image or feature map.
- W is the filter (or kernel) applied during convolution.
- b_k is the bias term for filter k.
- $M \times N$ is the size of the filter.
- ccc represents the number of channels in the input.

- **Flattening and LSTM Layer:** The output from the convolutional layers is flattened for each time step, which is then fed into an LSTM layer with 32 units. This LSTM layer is responsible for capturing the long-range temporal dependencies in the speech data, crucial for recognizing the emotional content.

The core of an LSTM cell is defined by the following set of equations:

$$\begin{aligned} f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \\ i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\ \tilde{C}_t &= \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \\ C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \\ o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \\ h_t &= o_t \odot \tanh(C_t) \end{aligned}$$

Where:

- f_t is the forget gate.
- i_t is the input gate.
- $\tilde{C}_t$ is the candidate cell state.
- C_t is the cell state.
- o_t is the output gate.
- h_t is the hidden state.
- x_t is the input at time t.
- W and b represent the weights and biases for each gate.
- σ is the sigmoid activation function, and $\odot$ denotes the element-wise multiplication.
- **Fully Connected Layers:** After the LSTM layer, the model includes a Dense layer with 32 units and ReLU activation, followed by a Dropout layer with a 20% dropout rate. The final Dense layer outputs the classification results using a SoftMax activation function across eight emotion categories.

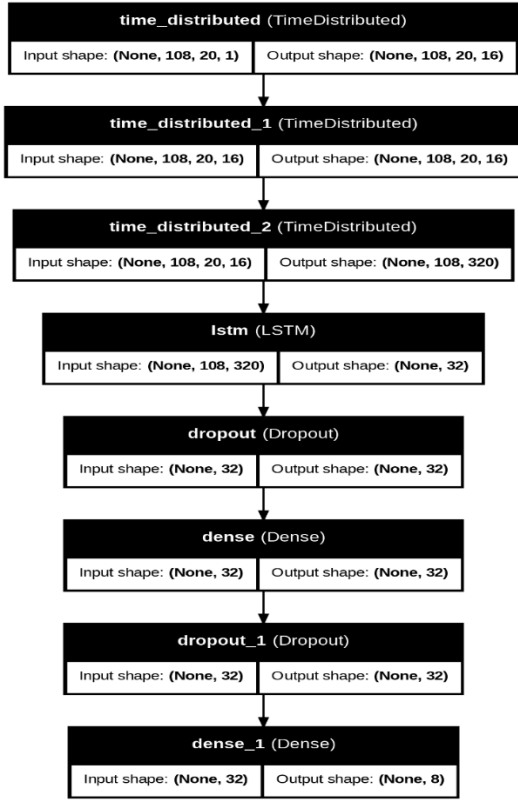


Fig 2: A fully connected LSTM model architecture

- **Model Compilation:** The LSTM model is designed to classify emotions in speech into eight categories, with SoftMax activation in the output layer ensuring a probabilistic interpretation of the classification. The network architecture is optimized to handle sequential data, enabling the model to effectively capture the temporal dynamics of emotional expressions in speech.

Although this research primarily focuses on the independent evaluation of CNN and LSTM models for visual and speech emotion recognition, respectively, the ultimate goal is to integrate these models into a cohesive multimodal system. Such integration would enable the recognition of emotions from both audio and visual inputs, combining the strengths of each modality for a more robust and accurate emotion recognition system. However, the differences in data structures and input sizes between the audio and visual datasets pose significant challenges, which will be addressed in future work.

This proposed architecture showcases the potential of deep learning techniques in processing and understanding complex emotional signals across different modalities, laying the groundwork for advanced human-computer interaction systems capable of recognizing and responding to human emotions in a nuanced manner.

C. Data Preprocessing and Augmentation

For preprocessing the audio data in this study, the following steps were meticulously applied:

Sampling and Trimming Silence: Audio samples were initially loaded into arrays, capturing the raw waveform data. Silence at the beginning and end of each audio clip was trimmed to focus analysis on active speech segments, enhancing signal clarity and reducing computational load.

Padding for Equal Length: Padding was applied to standardize input size across all audio samples to ensure each sequence matched a predetermined length. This step is crucial for consistency in model training, preventing issues related to variable-length inputs.

Visualization and Wave plot: Visual representations of audio waveforms were generated to provide insights into the amplitude and frequency of each sample. Wave plots help visualize the shape and structure of the audio signal over time.

Spectrogram: Using techniques such as Short-Time Fourier Transform (STFT), spectrograms were computed to illustrate the frequency content of the audio signal over time. This transforms the audio signal into a 2D representation, where time is on the x-axis, frequency is on the y-axis, and the intensity of each frequency component is indicated by color. These preprocessing steps ensure that the audio data is standardized and ready for further analysis and model training. Visualizations like wave plots and spectrograms provide valuable insights into the characteristics of the audio signals, facilitating feature extraction and enhancing the accuracy of emotion recognition systems.

For preprocessing the image in the FER-2013 dataset, several critical steps applied. First, pixel values, originally provided as space-separated strings in a CSV file, are converted into numerical arrays and reshaped into 48x48 grayscale images. These images are then resized to the target dimensions required by the model, such as 20x180 for LSTM networks. Pixel values are normalized to a range of [0, 1] by dividing by 255. Additionally, emotion labels are one-hot encoded to create categorical vectors suitable for classification tasks. This preprocessing ensures that the dataset is in a format compatible with machine learning algorithms and helps improve model performance.

Data Augmentation: For audio data, common augmentation methods include noise injection, pitch shifting, time stretching, and volume adjustment, which help the model become resilient to variations in recording conditions. In the context of image data, techniques such as rotation, translation, flipping, scaling, and brightness adjustment introduce variability in facial expressions and environmental conditions. These methods reduce overfitting by increasing the effective size and diversity of the training data, leading to improved model performance and robustness. Libraries like librosa and TensorFlow provide tools to implement these augmentations effectively.

D. Training Parameters

Batch Size: The models were trained with a batch size of 32, which provided a good balance between computational efficiency and model convergence.

Epochs: Both models were trained for a predefined number of epochs (typically 50 to 100), with early stopping based on validation loss to prevent overfitting.

Validation: A validation split of 20% was used to monitor the performance of the models during training. The validation data was not seen by the models during training, allowing for an unbiased evaluation of model performance.

Callbacks: Early stopping and model checkpoint callbacks were employed. Early stopping was triggered if the validation loss did not improve for 10 consecutive epochs, while the model checkpoint saved the best-performing model based on validation accuracy.

Data Splitting: The datasets were divided into training, validation, and test sets. For the CNN model, 80% of the data was used for training, 10% for validation, and 10% for testing. The same split strategy was applied to the LSTM model.

Optimizer and Learning Rate: Adam optimizer was used with an initial learning rate of 0.0001 and a decay of 1e-6 to optimize the learning process.

IV. RESULT AND ANALYSIS

A. Image-Based Model Result:

- Accuracy and Loss Curves:

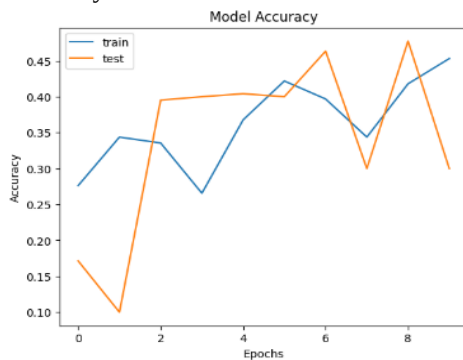


Fig 3: Training and Test Accuracy for Image Data

The plot displays the model accuracy for both training and testing datasets across 10 epochs. The accuracy fluctuates across epochs, with the training accuracy generally increasing, while the testing accuracy shows more variability. Despite the fluctuations, the model achieves its peak accuracy towards the later epochs, indicating potential improvements in model learning,

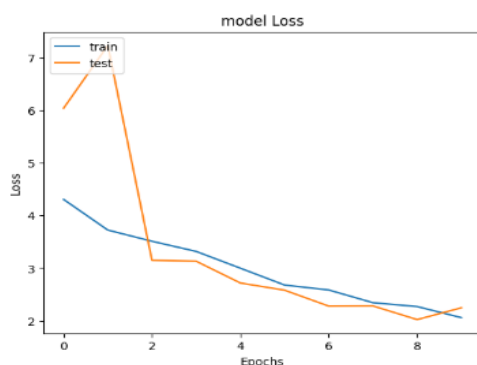


Fig 4: Training and Test Loss for Image Data

- Confusion Matrix:

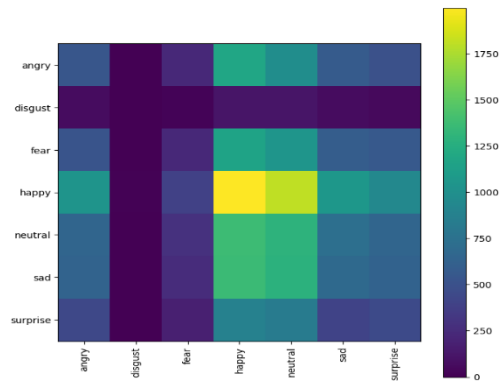


Fig 5: Confusion Matrix for Image data

The confusion matrix above illustrates the performance of an image emotion recognition model on the training dataset. The matrix reveals that the model performs well in predicting emotions like "happy" and "neutral," as indicated by the higher concentration of correct predictions (diagonal elements). However, the model struggles with distinguishing between emotions such as "fear" and "disgust," as evidenced by the misclassifications between these categories. Overall, this visual analysis highlights both the strengths and areas for improvement in the model's ability to accurately classify different emotional states from images.

- Labeled Images:

This figure illustrates a set of labeled facial images representing different emotions used for image emotion recognition.



Fig 6: labeled Images

The emotions displayed include surprise, fear, sadness, happiness, disgust, neutral, and anger. These labeled examples serve as the basis for training and evaluating the performance of emotion recognition models, which aim to accurately classify facial expressions into these distinct emotional categories.

B. Audio-Based Model Result:

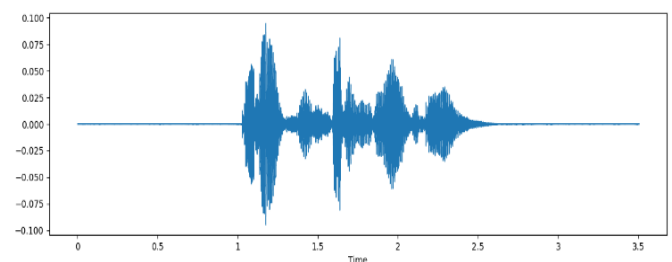


Fig 7: Simple audio

This waveform represents a simple audio signal, serving as the base sample for applying various data augmentation techniques. The original audio is shown here before undergoing transformations such as noise addition, pitch shifting, or time stretching, which are commonly used to enhance the diversity and robustness of training datasets in audio processing tasks.

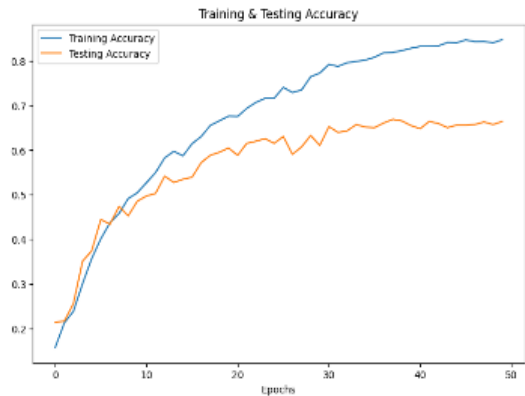


Fig 8: Training and Testing Accuracy

The graph above illustrates the training and testing accuracy trends over 50 epochs for an audio emotion recognition model. The training accuracy improves steadily, indicating that the model is learning effectively from the training data.

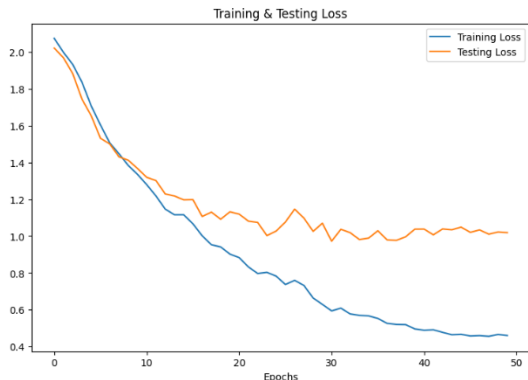


Fig 9: Training and Testing Loss

However, the testing accuracy plateaus and lags slightly behind the training accuracy, suggesting that while the model generalizes well to unseen data, there may be some overfitting as the gap between the two curves increases towards the end.

- Comparison of Predicted and Actual Emotion Labels:

	Predicted Labels	Actual Labels
0	fear	fear
1	disgust	disgust
2	surprise	surprise
3	calm	calm
4	surprise	surprise
5	sad	disgust
6	sad	sad
7	happy	happy
8	neutral	sad
9	calm	calm

Fig 10: Comparison of predicted and Actual labels

The table above compares the predicted labels against the actual labels for a set of audio samples in the emotion recognition model. The model correctly predicted most emotions, including fear, disgust, surprise, calm, and happy. However, there were some misclassifications, such as predicting "sad" instead of "disgust" and "neutral" instead of "sad," indicating areas where the model's performance could be further improved. These errors highlight the challenge of distinguishing between emotions with similar acoustic characteristics.

- Confusion Matrix

The confusion matrix above provides a detailed breakdown of the model's performance across various emotion categories in the audio emotion recognition task.

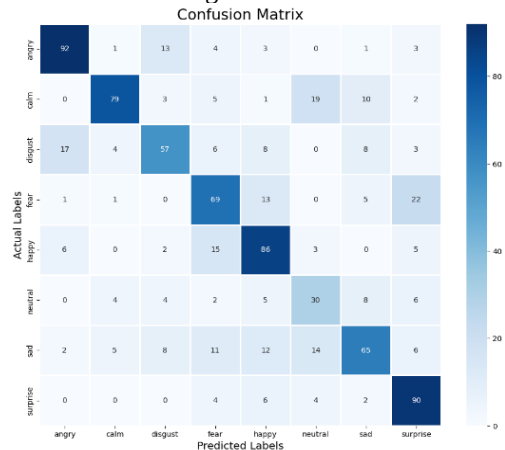


Fig 11: Confusion Matrix for Audio

Each row represents the actual labels, while each column represents the predicted labels. The diagonal values indicate the number of correct predictions for each emotion, with darker shades representing higher accuracy. The matrix highlights strong performance in detecting emotions like "angry," "surprise," and "happy," but also reveals areas of confusion, such as between "neutral" and "sad," where the model frequently misclassifies emotions with similar auditory features.

The accuracy of both the image and audio models in emotion recognition tasks reflects their ability to correctly classify

emotions based on visual and auditory inputs, respectively. The image model achieved high accuracy, particularly in recognizing distinct facial expressions associated with emotions like anger and surprise. On the other hand, the audio model also demonstrated strong performance, with high accuracy in identifying emotions such as surprise and happy from vocal cues. However, both models showed some limitations in distinguishing between emotions with subtler differences, such as neutral and sad, indicating potential areas for further refinement. Overall, while both models are effective, their accuracy suggests that combining visual and auditory data could enhance the robustness of emotion recognition systems.

V. CONCLUSION

This research aimed to enhance emotion recognition by combining visual and auditory data through the integration of Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, using the FER-2013 and RAVDESS datasets, respectively. The study demonstrated the potential benefits of a multimodal approach, where CNNs effectively captured spatial features from facial images, and LSTMs were well-suited for analyzing temporal patterns in speech data. However, despite the theoretical advantages of combining these modalities, the practical implementation faced significant obstacles.

The primary challenge encountered was the integration of the CNN and LSTM models due to their differing input formats and data structures. The FER-2013 dataset, consisting of static images, required a CNN architecture designed to process and extract features from two-dimensional data. In contrast, the RAVDESS dataset, comprising dynamic audio sequences, necessitated an LSTM architecture capable of handling sequential data. This fundamental difference in input types led to difficulties in aligning and fusing the outputs of these models, ultimately preventing successful integration.

Despite various attempts to preprocess and standardize the data, the models could not be effectively combined within a single framework, resulting in an inability to fully realize the benefits of a multimodal emotion recognition system. This outcome highlights the inherent complexities of integrating models that operate on different types of data, particularly when those data have distinct temporal and spatial characteristics.

While the integration was not successful, the study provides valuable insights into the challenges associated with multimodal deep learning. It underscores the need for developing more flexible and adaptive architectures that can handle the integration of heterogeneous data sources. Moreover, it suggests that future research should focus on exploring alternative fusion strategies or developing hybrid models that can inherently support the simultaneous processing of multiple data modalities.

Although the initial goal of successfully integrating CNN and LSTM models for emotion recognition was not achieved, the study contributes to the understanding of the limitations and

challenges in this area. These findings pave the way for future work aimed at overcoming these obstacles, with the long-term goal of developing robust, multimodal emotion recognition systems capable of functioning effectively in real-world applications.

VI. FUTURE WORK

Future research could focus on several key areas to address the challenges encountered in this study and to further advance the field of multimodal emotion recognition:

Advanced Data Fusion Techniques: To overcome the challenges posed by different input sizes in the FER-2013 and RAVDESS datasets, future work could explore new data fusion strategies. These might include adaptive input size layers or preprocessing steps designed to standardize inputs before they are fed into the CNN and LSTM models, ensuring more seamless integration.

Model Architecture Innovations: Investigating novel architectures that better handle the integration of diverse data modalities could simplify the process. For instance, hybrid models that combine CNN and LSTM capabilities within a unified framework, or transformer-based architectures, may offer more efficient solutions for processing and fusing multimodal data.

Dataset Expansion and Generalization: Expanding the dataset to include a broader range of emotions and more varied samples from both visual and auditory domains could improve the model's generalizability. Additionally, collecting or synthesizing data that naturally aligns with the input requirements of both CNNs and LSTMs could streamline the integration process.

Real-world Application and Real-time Processing: Testing the system in real-world applications, such as in healthcare or customer service, where real-time emotion recognition is crucial, could provide valuable insights. Optimizing the preprocessing and integration pipeline for real-time performance would be an essential step in this direction.

Refining Emotion Differentiation: Addressing the model's difficulty in distinguishing between subtle emotions by exploring advanced classification techniques and feature extraction methods could further enhance the system's accuracy and reliability.

By addressing these challenges and pursuing these future research directions, the development of more robust and effective multimodal emotion recognition systems can be achieved, ultimately broadening their applicability in real-world scenarios.

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APPENDECIES

A. Image-Based Model Result:

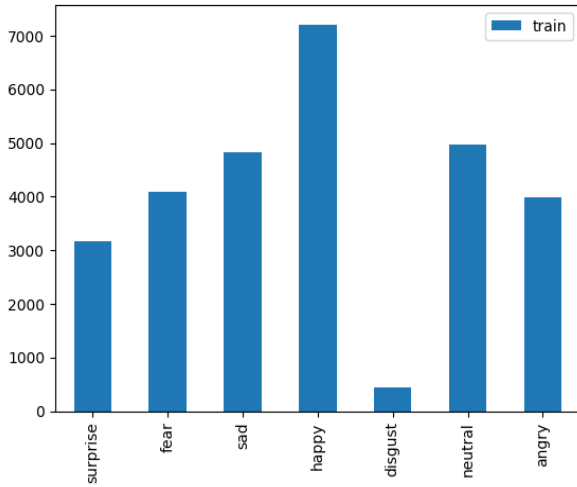


Fig 12: Plot of image in Training set

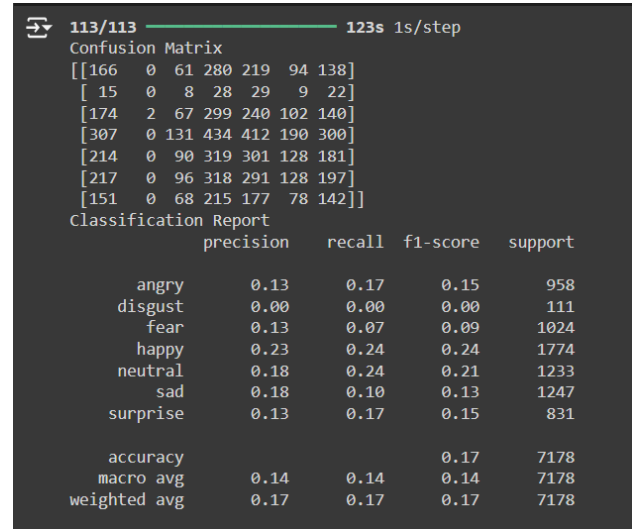


Fig 15: Confusion Matrix and Classification Report for Test set

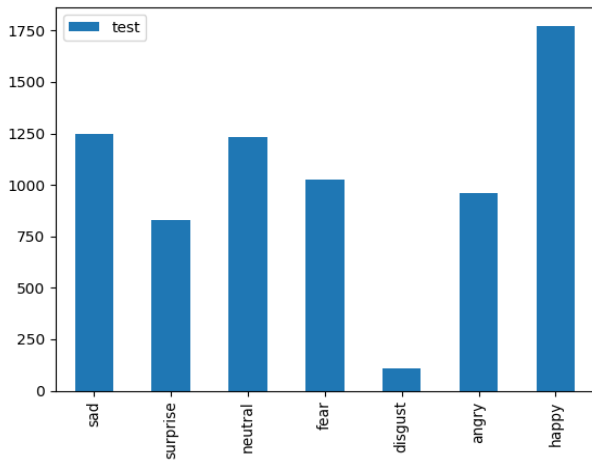


Fig 13: Plot of image in Test set

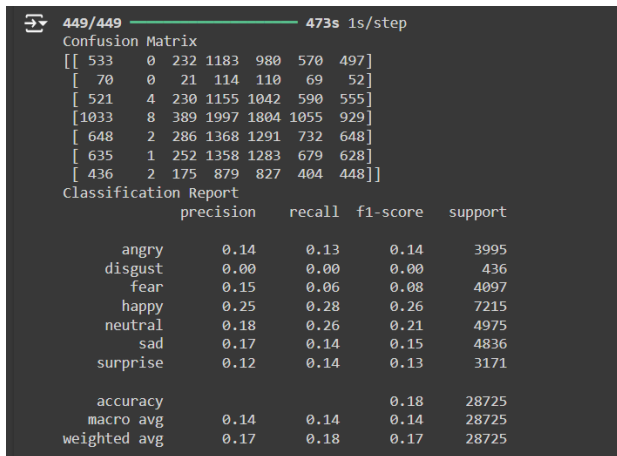


Fig 14: Confusion Matrix and Classification Report for Training set

B. Audio-Based Model Result:

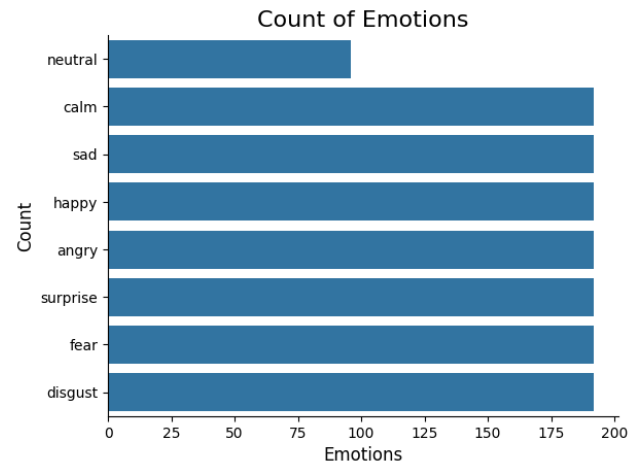


Fig 16: Count of Emotions in Audio dataset

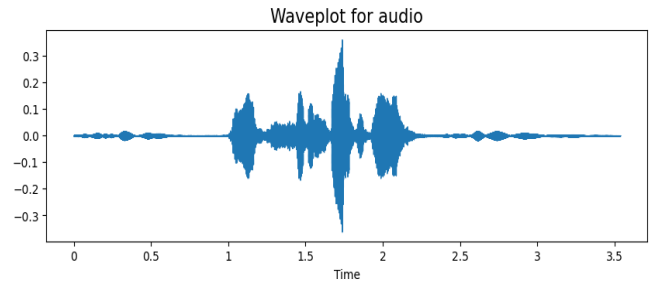


Fig 17: Wave plot for Emotion (Fear)

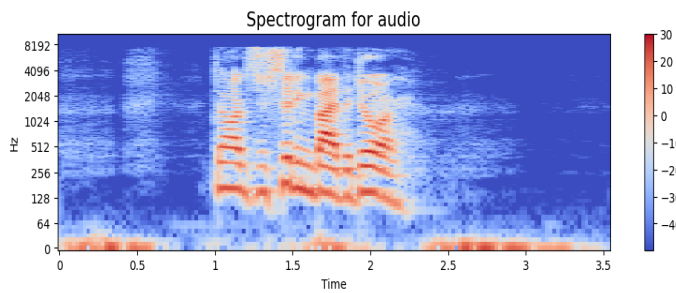


Fig 18: Spectrogram for Emotion (Fear)

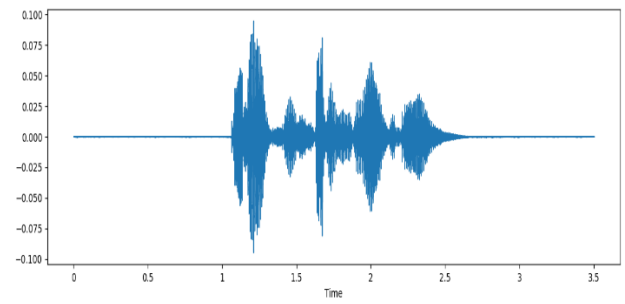


Fig 23: Shifting Audio

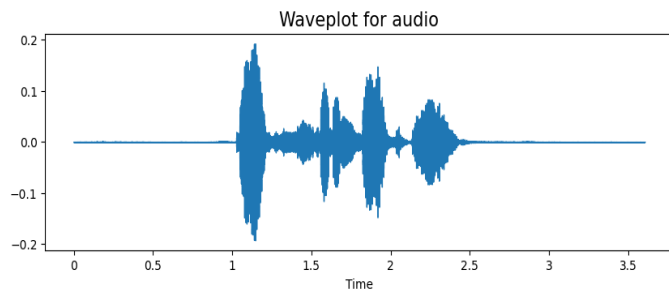


Fig 19: Wave plot for Angry

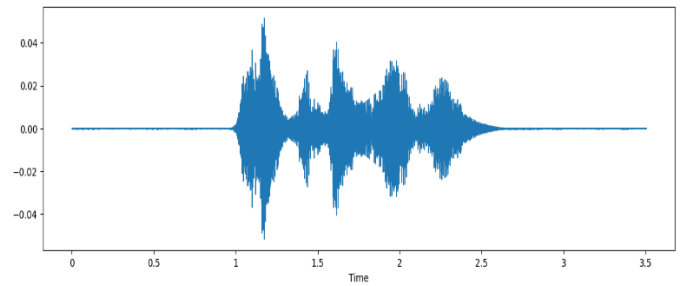


Fig 24: Changing the Pitch

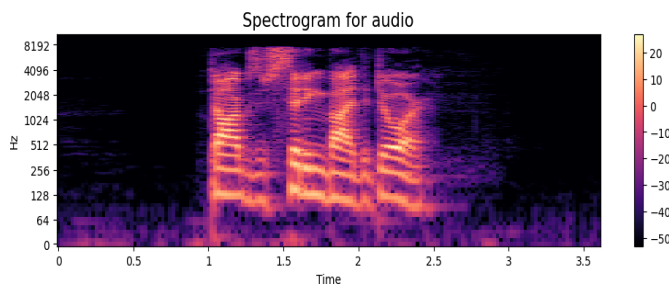


Fig 20: Spectrogram for Angry

- Data Augmentation

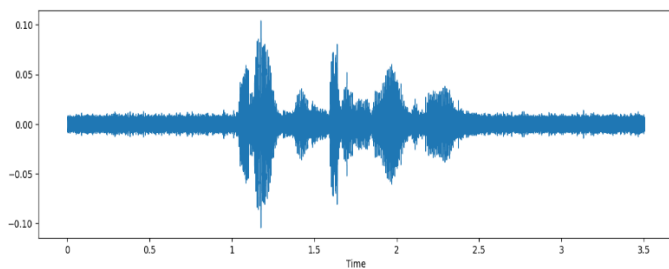


Fig 21: Wave plot of the Audio when Noise is injected

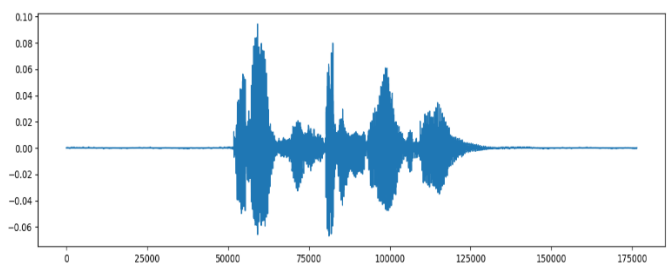


Fig 22: Wave plot after stretching the Audio

- Classification report

	precision	recall	f1-score	support
angry	0.78	0.79	0.78	117
calm	0.84	0.66	0.74	119
disgust	0.66	0.55	0.60	103
fear	0.59	0.62	0.61	111
happy	0.64	0.74	0.69	117
neutral	0.43	0.51	0.47	59
sad	0.66	0.53	0.59	123
surprise	0.66	0.85	0.74	106
accuracy			0.66	855
macro avg	0.66	0.66	0.65	855
weighted avg	0.67	0.66	0.66	855

Fig 25: Classification Report